

Methods	Train Acc	Test Acc (FF)	Test Acc (BA)	Test Acc (CM)	Test Acc (NV)
SNAIL [19]	59.2 $\pm$ 1.0	56.3 $\pm$ 3.5	60.2 $\pm$ 3.6	60.1 $\pm$ 3.1	61.3 $\pm$ 0.8
ProtoNet [20]	73.3 $\pm$ 0.2	64.6 $\pm$ 0.9	72.4 $\pm$ 0.8	62.4 $\pm$ 1.3	65.4 $\pm$ 1.2
MetaOptNet [21]	75.9 $\pm$ 0.4	60.3 $\pm$ 0.6	71.7 $\pm$ 2.5	61.7 $\pm$ 1.1	63.3 $\pm$ 1.9
ANIL [22]	69.7 $\pm$ 0.9	56.6 $\pm$ 1.0	59.0 $\pm$ 2.0	59.6 $\pm$ 1.3	61.0 $\pm$ 1.5
Meta-Baseline-SC [23]	75.4 $\pm$ 1.0	66.3 $\pm$ 0.6	73.3 $\pm$ 1.3	63.5 $\pm$ 0.3	63.9 $\pm$ 0.8
Meta-Baseline-MoCo [23]	81.2 $\pm$ 0.1	65.9 $\pm$ 1.4	72.2 $\pm$ 0.8	63.9 $\pm$ 0.8	64.7 $\pm$ 0.3
WReN-Bongard [17]	78.7 $\pm$ 0.7	50.1 $\pm$ 0.1	50.9 $\pm$ 0.5	53.8 $\pm$ 1.0	54.3 $\pm$ 0.6
CNN-Baseline	61.4 $\pm$ 0.8	51.9 $\pm$ 0.5	56.6 $\pm$ 2.9	53.6 $\pm$ 2.0	57.6 $\pm$ 0.7
Human (Expert)	-	92.1 $\pm$ 7.0	99.3 $\pm$ 1.9	90.7 $\pm$ 6.1	
Human (Amateur)	-	88.0 $\pm$ 7.6	90.0 $\pm$ 11.7	71.0 $\pm$ 9.6	

Table 1: Model performance versus human performance in BONGARD-LOGO. We report the test accuracy (%) on different dataset splits, including free-form shape test set (FF), basic shape test set (BA), combinatorial abstract shape test set (CM), and novel abstract shape test set (NV). Note that for human evaluation, we report the separate results across two groups of human subjects: *Human (Expert)* who well understand and carefully follow the instructions, and *Human (Amateur)* who quickly skim the instructions or do not follow them at all. The chance performance is 50%.

the 20 novel combinations are exclusive in the test set. The rationale behind this test set is that understanding the abstract shape concepts requires a model’s *abstraction* ability, as it is challenging to conceptualize abstract meanings from large shape variations.

Novel abstract shape test set This test set (NV) includes 320 abstract shape problems. Our goal here is to evaluate the ability of a model on the *discovery* of new abstract concepts. Different from the construction of the combinatorial abstract shape test set (CM), we hold out one attribute and all its combinations with other attributes from the training set. All problems related to the held-out attribute are exclusive in this test set. Specifically, we choose "have_eight_straight_lines" as the held-out attribute, since it presumably requires minimal effort for the model to extrapolate given that other similar "have_[xxx]_straight_lines" attributes already exist in the training set.

3.3 Quantitative Results

We report the test accuracy (Acc) of different methods on each of the four test sets respectively, and compare the results to the human performance in Table 1. We put the experiment setup for training these methods to Appendix C, and the results are averaged across three different runs. To show the human performance on our benchmark, we choose 12 human subjects to test on randomly sampled 20 problems from each test set. Note that for human evaluation, we do not differentiate test set (CM) and test set (NV) and thus report only one score on them, as humans essentially perform the same kind of new abstract discovery on both of the two test sets. To familiarize the human subjects with BONGARD-LOGO problems, we describe each problem type and provide detailed instructions on how to solve these problems. It normally takes 30-60 minutes for human subjects to fully digest the instructions. Depending on the total time that a human subject spends on digesting instructions and completing all tasks, we have split 12 human subjects into two evenly distributed groups: *Human (Expert)* who well understand and carefully follow the instructions, and *Human (Amateur)* who quickly skim the instructions or do not follow them at all.

Performance analysis In Table 1, we can see that there exists a significant gap between the Human (Expert) performance and the best model performance across all different test sets. Specifically, Human (Expert) can easily achieve nearly perfect performances (>99% test accuracy) on the basic shape (BA) test set, while the best performing models only achieve around 70% accuracy. On the free-form shape (FF), combinatorial abstract shape (CM) and novel abstract shape (NV) test sets where the existence of infinite vocabulary or abstract attributes makes these problems more challenging, Human (Expert) can still get high performances (>90% test accuracy) while all the models only have around or less than 65% accuracy. The considerable gap between these SOTA learning approaches and human performance implies these models fail to capture the core human cognition properties, as mentioned in Section 2.2.

Comparing Human (Expert) and Human (Amateur), we can see Human (Expert) always achieve better test accuracies with lower variances. This advantage becomes much more significant in the abstract shape problems, confirming different levels of understanding of these abstract concepts among human subjects. Comparing different types of methods, meta-learning methods that have been specifically

designed for few-shot classification have a greater potential to address the BONGARD-LOGO tasks than the non-meta-learning baselines (i.e., WReN-Bongard and CNN-Baseline). In Table 1, we can see the better test accuracies of most meta-learning models across all the test sets, compared with the non-meta-learning models. Interestingly, WReN, the model for solving RPMs [17], suffers from the severest overfitting issues, where its training accuracy is around 78% but its test accuracies are only marginally better than random guess (50%). This manifests that the ability of learning to learn is crucial for generalizing well to new concepts.

Furthermore, we perform an ablation study on BONGARD-LOGO, where we train and evaluate on the subset of 12,000 free-form shape problems in the same way as before. As the properties of *context-dependent perception* and *analogy-making perception* are not strongly presented in these free-form problems, concept learning on this subset has a closer resemblance to standard few-shot visual recognition problems [4, 33]. We thus expect a visible improvement in their performances. This is confirmed by the results in Table 2 in Appendix C, where almost all the methods achieve better training and test performances. Specifically, the best training accuracy of methods increases from 81.2% to 96.4%, and the best test accuracy (FF) of methods increases from 66.3% to 74.5%. However, there still exists a large gap between the model and human performance on free-form shape problems alone (74.5% vs. 92.1%). It implies the property of *few-shot learning with infinite vocabulary* has already been challenging for these methods.

We then compare different meta-learning methods and diagnose their respective failure modes. First, Meta-Baseline-MoCo performs best on the training set and also achieves competitive or better results on most of the test sets. It confirms the observations in prior work that good representations play an integral role in the effectiveness of meta-learning [23, 28]. Between the two Meta-Baselines, we can see that the MoCo pre-training is more effective in improving the training accuracy but the improvements become marginal on the test sets. This implies that the SOTA unsupervised pre-training methods improve the overall performance while still facing severe overfitting issues. Second, we perform another ablation study on model sizes. The results in Figure 5 in Appendix C show that the generalization performances of different models generally get worse as model size decreases.

Besides, most meta-learning models perform much better on the basic shape problems than on the abstract shape problems. This phenomenon is consistent with the human experience, as it takes a much longer time for humans to finish a latter task. How to improve the combinatorial generalization or extrapolation of abstract concepts in our benchmark remains a challenge for these models. We also observe that each type of meta-learning methods has its preferred generalization tasks, according to their different behaviors on test sets. For example, the memory-based method (i.e., SNAIL) seems to try its best to perform well on abstract shape problems by sacrificing its performance in the free-form shape and basic shape problems. The metric-based method (i.e., ProtoNet) performs similarly to Meta-Baselines and better than memory-based and optimization-based methods. Lastly, the optimization-based methods (i.e., MetaOptNet and ANIL) tend to have a larger gap between training and test accuracies. These distinguishable behaviors on our benchmark provide a way of quantifying the potential advantages of a meta-learning method in different use cases.

4 Related Work

Few-shot learning and meta-learning The goal of few-shot learning is to learn a new task (e.g., recognizing new object categories) from a small amount of training data. Pioneer works have approached it with Bayesian inference [33] and metric learning [34]. A rising trend is to formulate few-shot learning as meta-learning [4, 5]. These methods can be categorized into three families: 1) memory-based methods, e.g., a variant of MANN [30] and SNAIL [19], 2) metric-based methods, e.g., Matching Networks [35] and ProtoNet [20], and 3) optimization-based methods, e.g., MAML [5], MetaOptNet [21] and ANIL [22]. Recent work [23, 28] has achieved competitive or even better performances on few-shot image recognition benchmarks [3, 36] with a simple pre-training baseline than advanced meta-learning algorithms. It also sheds light on a rethinking of few-shot image classification benchmarks and the associated role of meta-learning algorithms.

Concept learning and Bongard problem solvers Concept learning methods concern about transforming sensory observations [37] and experiences [38] into abstract concepts, which serve as the building blocks of understanding and reasoning. A rich and structured representation of concepts is programs [3, 25] that model the generative process of observed data in an analysis-by-synthesis framework. Program-based (or rule-based) methods have been applied to tackle BPs [12, 39, 13].

RF4 [12] is an ILP system where the images are hand-coded into logical formulas. Phaeaco [39] is an exemplar BP solver that implements the FARG concept architecture [10]. [13] proposes to translate visual features into a formal language with which the solver applies Bayesian inference for concept induction. However, these methods require a substantial amount of domain knowledge and do not offer complete solutions to original BPs, making them infeasible for our benchmark which consists of tens of thousands of problems. In contrast, [24] is the first work to show the usefulness of deep learning methods for solving BPs by using a pre-trained feature extracting neural network. All these results motivate the development of hybrid concept learning systems that integrate deep learning with symbolic reasoning, which have attracted much attention recently [40, 41]. The hybrid systems can offer great potential in combining the representational power of neural networks with the generalization power of symbolic operations for tackling our benchmark.

Abstract reasoning benchmarks In addition to popular machine perception benchmarks that focus on categorization and detection [1, 2], there have been previous efforts in creating new benchmarks for abstract reasoning, including compositional visual question answering [7], physical reasoning [42–44], mathematics problems [45], and general artificial intelligence [46]. Most relevant to our benchmark, the Raven-style Progressive Matrices (RPMs) [47], a well-known human IQ, have inspired researchers to design new abstract reasoning benchmarks [17, 48]. Our benchmark is complementary to RPMs: RPMs focus on relational concepts (such as progression, XOR, etc.) while BONGARD-LOGO problems focus on object concepts (such as stroke types, abstract attributes, etc.). In RPMs, the relational concepts come from a small set of five relations [17]. In our benchmark, the object concepts can vary arbitrarily with procedural generation. Recently, [49] has proposed V-RPM that extends RPMs to real images. As V-PRO is a real image version of RPMs, it differs from our benchmark in the similar ways described above. Therefore, in contrast to RPMs where automated pattern recognition models have achieved competitive performances [50, 17], our BONGARD-LOGO benchmark has posed a greater challenge to current models due to its fundamental shift towards more human-like (i.e., context-based, analogy-making, few-shot with infinite vocabulary) visual cognition.

5 Discussion and Future Work

We introduced a new visual cognition benchmark that emphasizes concept learning and reasoning. Our benchmark, named BONGARD-LOGO, is inspired by the original BPs [11] that were carefully designed in the 1960s for demonstrating the chasms between human visual cognition and computerized pattern recognition. In a similar vein as the original one hundred BPs, our benchmark aims for a new form of human-like perception that is context-dependent, analogical, and few-shot of infinite vocabulary. To fuel research towards new computational architectures that give rise to such human-like perception, we develop a program-guided problem generation technique that enables us to produce a large-scale dataset of 12K human-interpretable problems, making it digestible by data-driven learning methods to date. Our empirical evaluation of the state-of-the-art visual recognition methods has indicated a considerable gap between machine and human performance on this benchmark. It opens the door to many research questions: What types of inductive biases would benefit concept learning models? Are deep neural networks the ultimate key to human-like visual cognition? How is the Bongard-style visual reasoning connected to semantics and pragmatics in natural language?

Prior attempts on tackling the original BPs with symbolic reasoning [12] and program induction [13] have also been far away from solving BONGARD-LOGO. However, along with the inherent nature of compositionality and abstraction in BPs, they have supplied a great amount of insight on the path forward. Therefore, one promising direction is to develop computational approaches that integrate neural representations with symbolic operations in a hybrid system [8, 40], that acquires and reasons about abstract knowledge in a prolonged process [51], that establishes a tighter connection between visual perception and the high-level cognitive process [14, 3]. We invite the broad research community to explore these open questions together with us for future work.

Broader Impact

Our work created a new visual concept learning benchmark inspired by the Bongard problems. Our preliminary evaluations have illustrated a considerable gap between human cognition and machine recognition, highlighting the shortcomings of existing pattern recognition methods. In Machine Learning and Computer Vision, we have witnessed the integral role of standardized benchmarks [1, 2] in promoting the development of new AI algorithms. We would encourage future work to develop